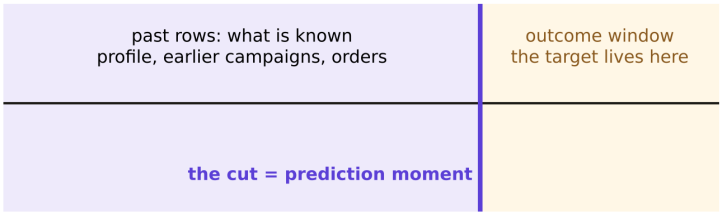
Cheat sheet: Week 5 Monday, regression and classification



Today's hard terms

Term	Plain meaning	Rule or formula	Trap
Query, rule, prediction	the answer exists; the decision is a condition; the future is scored	a model needs input now, target later, decision moves	"we have data, so it is ML"
Prediction time	the moment the score is made	inputs on or before it, target after it	counting the whole log
Target	what the model is scored against	repeat = a delivered order in the 90 days after the cutoff	a later order outside the window still counts as 0
Train, validation, test	learn, choose, report	test is spent once	a peeked test is validation now
Seed	the fixed shuffle	`random_state=42`, written down	trying seeds until the score is best
Baseline	the score of knowing nothing	majority class; the mean; a cheap rule	a 0.89 model beside a 0.887 baseline
Least squares	the line with the smallest squared misses	`slope = sum(dx*dy) / sum(dx^2)`	joining first and last point
Coefficient, linear	target units per input unit	Rs 120 per visit	read as a cause
Logistic regression	a line squeezed into 0 to 1	`p = 1 / (1 + exp(-(a + b x)))`	a line on 0/1 gives 1.28
Odds multiplier	strength of a logistic input	`exp(b)`; 1.05 gives 2.85	"adds b to the probability"
Residual	actual minus predicted	a pattern is a missed shape	a fit quoted outside its range

Three that stick

- Uber's DeepETA keeps the routing engine's ETA and predicts only how wrong it will be (uber.com blog, 10 February 2022, verified 28 October 2026).
- Google's Rules of ML, rule 1: "Don't be afraid to launch a product without machine learning" (verified 28 October 2026).
- At the bank, a one-line rule (call clients whose last campaign succeeded) scores 0.8988 on validation against the model's 0.8971.

When to use what

If the question is	Use	Unless
already answered in the data	a query	never a model
a decision someone can write down	a rule	its misses cost more than a model
"how much"	linear regression, beside the mean	the residuals bow: add the curve
"will they"	logistic regression, beside the majority class	a one-line rule ties it: ship the rule

The ML process, Monday lit



Everything so far, one line a day

Day	The one line	Day	The one line
W1 Mon	The revenue tree and the first honest numbers: refuse the average when one customer buys in bulk	W2 Fri	Excel for analysts: one number reaches the leadership deck
W1 Tue	Which lever moved: compare like with like windows and denominators	W3	Build 1: a Kalpa Health sub-problem scoped, built and defended
W1 Wed	Can we trust the numbers: profile, clean, reconcile, recompute	W4 Mon	Metric design: the number the plan chases, and what stops it being gamed
W1 Thu	Real or noise, cause or coincidence: the chance reference and the one-page note	W4 Tue	One slide, five minutes, three people paid to disagree
W1 Fri	The week rebuilt alone, and the note defended aloud	W4 Wed	Market basket: which pairs lift frequency, and what each is worth
W2 Mon	The revenue tree as queries the warehouse runs every Monday	W4 Thu	Cohort and funnel: where the loss actually happens
W2 Tue	Booked against collected: joining payments without lying	W4 Fri	Forecasting against a baseline that is hard to beat
W2 Wed	Window functions: top members, falling spend, the running total against plan	**W5 Mon**	**Frame it, fix the prediction time, seal the test, beat the baseline, read both models, read the residuals**
W2 Thu	pandas groupby, merge and reshape: the customer table Marketing refreshes		